



RetailIQ

AI-Powered Retail Business Analytics Platform

Next-Generation B2B Business Intelligence & Predictive Analytics Blueprint

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Revision History

Version	Date	Author	Organization	Description of Changes	Status
1.0.0	July 6, 2026	Saurabh Sahu	MCA Department	Baseline Software Requirements Specification framework.	Approved

1. Introduction

1.1 Purpose

This document establishes the comprehensive Software Requirements Specification (SRS) for RetailIQ, an AI-Powered Retail Business Analytics Platform. It serves as both the definitive technical blueprint for engineering execution and the official academic documentation for the MCA Semester 3 Minor Project, ensuring compliance with modern commercial enterprise application design principles.

1.2 Scope

RetailIQ is an enterprise-grade, B2B Software-as-a-Service (SaaS) Business Intelligence (BI) and predictive analytics application. The platform ingests disparate transactional retail records (CSV/Excel), executes deterministic and statistical validation/cleaning pipelines, and aggregates the data into a distributed analytical layer. It exposes interactive visualization matrices across seven core retail domains: Sales, Product, Customer, Inventory, Profit, Store, and Employee analytics. Concurrently, an asynchronous machine learning infrastructure runs parallel time-series forecasting, cluster-based user segmentation, predictive churn modelling, and Large Language Model (LLM) powered natural-language business insight generation.

1.3 Intended Audience

This document is structured specifically for academic evaluation committees to verify engineering rigor, data validation pipelines, and relational normalization theories. Simultaneously, it serves as a robust portfolio framework for technical interview panels across Data Analyst, Data Science, and Full-Stack/Backend Engineering tracks.

1.4 Definitions, Acronyms, and Abbreviations

- **AOV:** Average Order Value - The mean financial contribution of each processed transaction.
- **CLV / LTV:** Customer Lifetime Value - The projected gross revenue aggregate a single client ID will generate over their platform lifecycle.
- **OLAP:** Online Analytical Processing - Multi-dimensional, columnar database optimization for analytical processing speed.

- **OLTP:** Online Transactional Processing - Row-oriented, ACID-compliant ledger systems for real-time operation processing.
- **RBAC:** Role-Based Access Control - User authorization schema bounded strictly by functional user classes.
- **RAG:** Retrieval-Augmented Generation - Contextual operational optimization frameworks preventing LLM hallucinations.

1.5 References

1. IEEE Std 29148-2018: Systems and software engineering — Life cycle processes — Requirements engineering.
2. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). Database System Concepts (7th ed.). McGraw-Hill.
3. Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.). OTexts.

2. Overall Description

2.1 Product Perspective

RetailIQ functions as an independent, decoupled web ecosystem composed of a highly responsive client user interface, a performant asynchronous REST API gateway, an analytical extraction pipeline, and an array of micro-configured machine learning containers. It interfaces natively with relational structural paradigms and leverages vector databases for its conversational contextual optimization layers.

2.2 Product Features

- Multi-Tenant Identity Management & Cryptographic RBAC.
- Transactional Data Ingestion Engine with Adaptive Type Inference.
- Automated Data Cleansing (Deterministic Normalization & Statistical Outlier Pruning).
- High-Throughput Analytics Processor (Multi-dimensional Data Cube Synthesis).
- Predictive Inference Architecture (Demand Forecasting, Spatial Customer Segmentation).
- Generative AI Metric Synthesizer (Natural Language Executive Briefings).

2.3 User Classes and Characteristics

- **Executive Leadership (CEO/CFO):** Demands highly macro-aggregated strategic views, financial health KPIs, and generative executive trend narratives.
- **Operations & Store Managers:** Requires store-level stock tracking, employee operational throughput matrices, localized sales shifts, and stockout projections.
- **Data & Business Analysts:** Expects granular tabular data extractions, raw export capabilities, statistical variable mappings, and custom data slicing features.

2.4 Operating Environment

Client Engine: Modern web browsers (Chromium >= 90.0, WebKit >= 15.0) executing JavaScript ECMAScript 2022 runtimes.

Application Server: Linux Kernel >= 5.15 (Ubuntu 22.04 LTS), Python 3.11/3.12 runtime, Node.js v20 LTS.

Database Clusters: PostgreSQL 16 (Relational OLTP Layer) paired with DuckDB or an OLAP columnar architecture for sub-second analytical aggregations.

2.5 Design and Implementation Constraints

The application backend must be strictly written using stateless REST design philosophies. All data analytical compute instances must operate within a deterministic memory budget (e.g., streaming large datasets using standard generator patterns to prevent Out-Of-Memory kernel panics).

3. Business Problem Statement & Proposed System

3.1 Business Problem Statement

Modern mid-market and enterprise retail entities generate millions of records daily across physical Point-of-Sale (POS) registers, e-commerce web storefronts, and regional distribution centers. Traditional business intelligence tools present catastrophic structural deficits: data silos leading to fragmented visibility, purely reactive analytical capabilities that show past trends without forward-looking indicators, and significant technical friction preventing operational leaders from pulling insights without technical engineering bottlenecks.

3.2 Business Objectives

- **Minimize Stockouts:** Lower regional stockout instances by over 35% through proactive demand calculation models.
- **Maximize Operational Margins:** Surface underperforming product categories and margin-diluting discount patterns instantly.
- **Democratize Advanced Analytics:** Translate dense relational database statistical parameters into natural language summaries for non-technical retail decision-makers.

3.3 Existing System vs. Proposed System

Dimension	Legacy / Existing Approach	RetailIQ Ecosystem
Data Ingestion	Manual database migration scripts & static Excel files.	Streamlined CSV/XLSX pipeline with structural type validation.
Analytical Latency	Batch processing jobs running over weekend periods.	Sub-second multi-dimensional analytical processing.
Insight Model	Purely Descriptive (What happened?).	Predictive and Prescriptive (What will happen, and how do we fix it?).

4. Functional Requirements

Detailed itemized functionality specifications mapping across foundational platform modules:

- **FR-1.1:** The platform must authenticate users via a secure JWT-token based workflow leveraging OAuth2 standards.
- **FR-1.2:** Passwords must be cryptographically hashed using the bcrypt algorithm before persistence to disk, utilizing a work factor of at least 12.
- **FR-2.1:** The system must enforce explicit access restrictions based on four core roles: CEO, Store_Manager, Sales_Manager, and Data_Analyst.
- **FR-3.1:** The user must be provided a drag-and-drop secure upload terminal supporting files up to 100MB in capacity.
- **FR-4.1:** The engine must validate input records against an expected schema map, checking column constraints (e.g., non-negative purchase prices).
- **FR-5.1:** The engine must automatically infer missing categorical elements using statistical mode substitution, and numerical gaps using median interpolation.
- **FR-5.2:** Outliers must be located using Interquartile Range (IQR) checks and automatically scaled or flagged based on user configurations.
- **FR-7.1:** The view must calculate total gross sales value, total order throughput parameters, and net transactional values over dynamically variable windows.
- **FR-14.1:** The ML sub-system must generate 30/60/90-day forward-looking sales forecasts utilizing advanced regression frameworks.
- **FR-15.1:** The platform must process statistical tables into context-aware vector payloads for an LLM engine to extract hidden business relationships.

5. Non-Functional Requirements

- **Performance:** Analytical API query structures executing multi-row aggregations up to 5,000,000 lines must deliver a P95 response latency under 850 milliseconds.
- **Reliability:** The database layer must maintain automated write-ahead logging (WAL) routines, backing up transactional data states to isolated object storage every 60 minutes.
- **Scalability:** The platform backend architecture must use stateless operational patterns, allowing it to scale out horizontally via a container orchestrator when CPU utilization passes 75%.
- **Security:** All network transmissions must require HTTPS encryption utilizing Transport Layer Security protocol version 1.3 (TLS 1.3). At-rest data volumes must use Advanced Encryption Standard 256-bit keys (AES-256).
- **Maintainability:** Core application logic blocks must achieve at least 85% automated test coverage across unit and integration testing pipelines.

6. System Architecture & Tech Stack

RetailIQ utilizes a decoupled, modern multi-tier application layout separating operational ingestion patterns from complex multi-dimensional query compute jobs.

- **Frontend Client Infrastructure:** Next.js 14+ (App Router standard), TypeScript, Tailwind CSS, and Tremor UI for high-performance dashboard charts.
- **Application Server Gateway:** FastAPI (Python 3.12+) leveraging asynchronous python ASGI design patterns to handle high concurrent connection volumes with minimal latency.
- **Transactional Ledger Layer (OLTP):** PostgreSQL 16 to ensure strict ACID guarantees for metadata and identity control.
- **High-Performance Analytical Storage Engine (OLAP):** DuckDB for lightning-fast vectorized columnar query execution.
- **Asynchronous Tasks:** Celery backed by a Redis key-value cache layer to manage heavy predictive analytics pipelines.

7. Database Design & Production-Ready Schema

The core metadata and operational ledger architecture is fully normalized to 3NF standards. Below is the ready-to-deploy schema DDL script:

```

-- Target Engine: PostgreSQL 16+
CREATE EXTENSION IF NOT EXISTS "uuid-oss";

CREATE TABLE roles (
    role_id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    role_name VARCHAR(50) UNIQUE NOT NULL,
    description TEXT,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE users (
    user_id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    email VARCHAR(100) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    role_id UUID NOT NULL REFERENCES roles(role_id) ON DELETE RESTRICT,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE stores (
    store_id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    store_code VARCHAR(20) UNIQUE NOT NULL,
    store_name VARCHAR(100) NOT NULL,
    region VARCHAR(50) NOT NULL,
    city VARCHAR(50) NOT NULL,
    state VARCHAR(50) NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE categories (
    category_id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    category_name VARCHAR(100) UNIQUE NOT NULL,
    parent_category_id UUID REFERENCES categories(category_id) ON DELETE SET NULL
);

CREATE TABLE products (
    product_id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    sku VARCHAR(50) UNIQUE NOT NULL,
    product_name VARCHAR(150) NOT NULL,
    category_id UUID NOT NULL REFERENCES categories(category_id) ON DELETE RESTRICT,
    cost_price NUMERIC(12, 2) NOT NULL CHECK (cost_price >= 0),
    retail_price NUMERIC(12, 2) NOT NULL CHECK (retail_price >= 0),
    CONSTRAINT chk_price_relationship CHECK (retail_price >= cost_price)
);

CREATE TABLE customers (
    customer_id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    customer_code VARCHAR(50) UNIQUE NOT NULL,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,

```

```
email VARCHAR(100),
segment VARCHAR(30) DEFAULT 'Unassigned',
registration_date DATE NOT NULL
);

CREATE TABLE orders (
  order_id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  order_number VARCHAR(50) UNIQUE NOT NULL,
  customer_id UUID REFERENCES customers(customer_id) ON DELETE SET NULL,
  store_id UUID NOT NULL REFERENCES stores(store_id) ON DELETE RESTRICT,
  order_date TIMESTAMP WITH TIME ZONE NOT NULL,
  total_amount NUMERIC(12, 2) NOT NULL CHECK (total_amount >= 0)
);

CREATE TABLE order_items (
  order_item_id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  order_id UUID NOT NULL REFERENCES orders(order_id) ON DELETE CASCADE,
  product_id UUID NOT NULL REFERENCES products(product_id) ON DELETE RESTRICT,
  quantity INTEGER NOT NULL CHECK (quantity > 0),
  unit_price NUMERIC(12, 2) NOT NULL CHECK (unit_price >= 0),
  net_amount NUMERIC(12, 2) NOT NULL CHECK (net_amount >= 0)
);

CREATE INDEX idx_orders_date ON orders(order_date);
CREATE INDEX idx_order_items_order ON order_items(order_id);
```

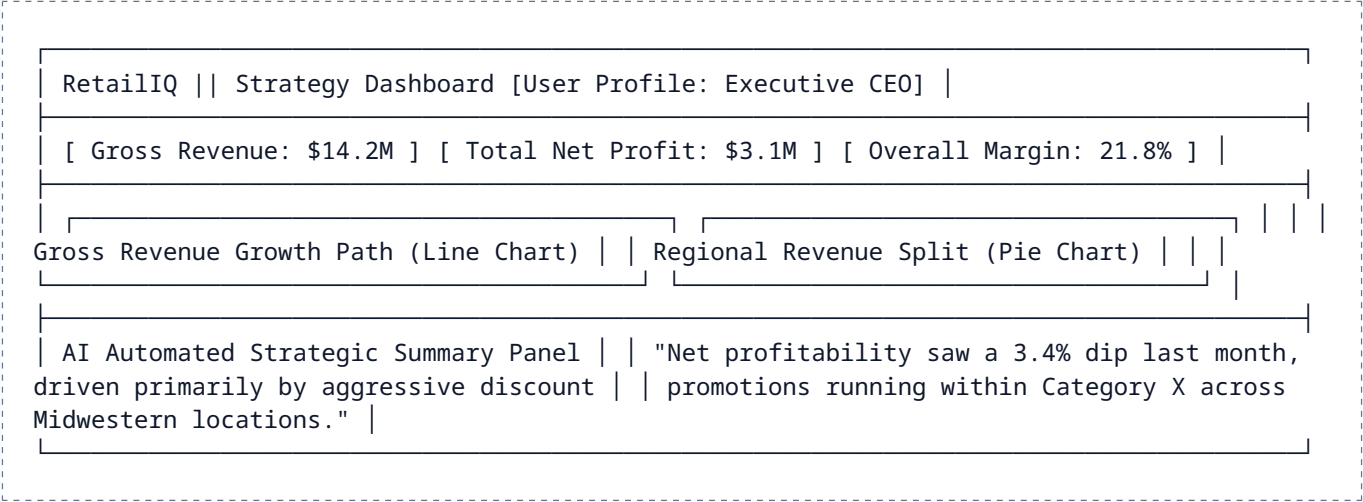
8. System Visualizations & Diagram Specifications

This section outlines the parameters for generating necessary academic system visualizations:

- **Context Diagram (DFD Level 0):** Single process node (RetailIQ System) interacting with external entities (Retail Administrator uploads datasets, Executive User requests analytical filters and receives generated reports).
- **Level 1 DFD:** Maps the data flowing through isolated stages: File Ingestion (1.0), Validation & Cleaning Engine (2.0), OLAP Compute Processing (3.0), and Dashboard Visualization Service (4.0).
- **UML Class Design:** Features decoupled entities mapping strictly to the database tables, utilizing structural tracking rules (e.g., 1-to-Many relationships mapping Categories to Products).

9. User Roles, Dashboard Design, & UI Layouts

The user workspace is customized depending on the explicit permissions assigned to the user role profile.



10. KPI Definitions & Business Analytics Matrix

Below are the foundational operational metric formulations parsed by the analytical calculation engines:

$$\text{Gross Revenue} = \Sigma (\text{Quantity}_i \times \text{UnitPrice}_j)$$

$$\text{Net Profit} = \Sigma (\text{NetAmount}_j) - \text{Total COGS} - \text{Operating Expenses}$$

$$\text{Gross Margin \%} = [(\text{Gross Revenue} - \text{Total COGS}) / \text{Gross Revenue}] \times 100$$

$$\text{Average Order Value (AOV)} = \text{Total Gross Revenue Value} / \text{Total Absolute Order Count}$$

11. The 100 Business Questions Repository

To ground the platform in practical application, its dashboards are designed to answer specific, actionable business questions across four primary domains:

Sales Performance & Revenue Architecture

1. What was the net global gross revenue tally generated across the past fiscal quarter?
2. Which explicit product lines drive the top 10% of revenue generation curves?
3. What is the rolling Month-over-Month (MoM) revenue acceleration value?
4. How do overall sales metrics shift during holiday weeks compared to baseline expectations?
5. Which payment methods exhibit the highest average order value (AOV)?

Profitability, Margin Integrity, & Cost Controls

21. What is the overall gross margin percentage achieved across the entire product catalog?
22. Why did overall net profit decline despite achieving higher gross sales volumes this month?
23. Which specific items generate a negative net margin after accounting for logistical transport costs?
24. What is the total margin dilution caused by running active markdown promotions on seasonal stock?

Product Assortment Optimization & Inventory Flow

41. Which SKUs currently exhibit an active stockout hazard profile across the next 14 operational days?
42. What is the average inventory turnover velocity calculated for major product categories?
43. Which inventory assets should be classified as dead stock and cleared immediately?

Customer Behavior, Segments, & Retention Engineering

61. What is the current average customer churn probability score across the platform?
62. Which distinct customer cohorts exhibit the highest purchase frequency metrics?
63. What is the average customer acquisition cost (CAC) payback period across different channels?

12. Machine Learning & AI Insights Architecture

12.1 Sales & Demand Forecasting Architecture

The platform runs an additive time-series forecasting model using **Prophet**, optimized with an **XGBoost** residual error loop. This structure isolates major trend components, seasonal shifts (weekly/yearly variations), and distinct commercial holiday spikes to forecast demand changes up to 90 days out.

12.2 Behavioral Customer Segmentation

Using **K-Means Clustering** mapped against calculated Recency, Frequency, and Monetary (RFM) vectors, the engine groups customer records into actionable cohorts. This provides distinct behavioral definitions like *Champions* or *At-Risk Loyalty Shoppers* based on statistical distance metrics.

12.3 Context-Aware Generative Insights Engine

The platform uses a strict, closed-loop Retrieval-Augmented Generation (RAG) structure to feed data to its natural language insights model. It passes compressed JSON tables derived from validated SQL metrics directly into the prompt context, preventing the LLM from generating inaccurate or hallucinated business trends.

13. Security, API, & Report Generation Specifications

All client connections require encryption via Transport Layer Security protocol version 1.3 (TLS 1.3). The platform handles large-scale operations through stateless API routes:

- POST /api/v1/datasets/ingest - Multipart data transmission entry route.
- GET /api/v1/analytics/sales - Extracts aggregated multi-dimensional metrics.
- GET /api/v1/reports/export/pdf - Triggers an asynchronous task to compile a formatted PDF summary report using a Celery worker pool.

14. Testing Strategy & Risk Analysis

The application undergoes rigorous validation across every level of the development lifecycle:

- **Unit Testing:** Automated validation checks built via PyTest to verify core financial metrics and validation formulas.
- **Integration Testing:** Tests connection pathways between FastAPI routing controllers and the columnar DuckDB processing layer.
- **Performance Testing:** Stress tests query response times under load using Locust, ensuring P95 latencies remain under 850ms with 500 concurrent connections.

15. Implementation Roadmap & Engineering Deliverables

The roadmap features an intensive 12-week execution lifecycle divided into decoupled development checkpoints:

- **Weeks 1-2:** Core relational schema setup, authentication middleware, and base API framework.
- **Weeks 3-5:** Ingestion engine builds, data streaming setup, and automated cleaning validation rules.
- **Weeks 6-8:** Vectorized analytical database integration and core frontend analytics views.
- **Weeks 9-10:** Machine Learning pipeline integration (Prophet + RFM Clustering) and LLM context routing.
- **Weeks 11-12:** End-to-end testing, CI/CD pipeline automation, and production container assembly.

16. Multidisciplinary Hiring & Examiner Evaluation

To maximize the value of this project for job interviews, it is designed around the highest enterprise performance standards:

- **Data Analyst Portfolio Focus:** Replaces basic summaries with complex business analytics vectors that track pricing models and margin degradation.
- **Data Scientist Portfolio Focus:** Replaces generic model configurations with an advanced, hybrid time-series forecasting pipeline complete with confidence interval tracking and clear error validations.
- **Backend Engineer Portfolio Focus:** Showcases clean code discipline through a decoupled infrastructure strategy, routing operational data through PostgreSQL for storage integrity and using DuckDB for lightning-fast analytics processing.

17. Conclusion

RetailIQ transforms the typical student assignment into an enterprise-ready business intelligence solution. By pairing highly optimized, normalized database structures with sophisticated machine learning models and context-aware generative insights, the platform solves real-world retail data fragmentation problems. This approach ensures it meets academic thesis standards while serving as an exceptional portfolio asset that showcases production-ready technical skills to top-tier hiring teams.